8.4 The Upper Bound Theorem

"What is the maximal number of k-faces for a d-polytope with n vertices?"

The answer to this question is the *upper bound theorem*: "The cyclic polytope $C_d(n)$ has the maximal number of k-faces for all k." This claim made by Motzkin [415] in 1957 became known as the *upper-bound conjecture*. During a long and involved history (see also Grünbaum [252]), including premature announcements and many partial results, it was proved for polytopes with "few" vertices (i.e., $n \leq d+3$, by Gale [222]) and for polytopes with "many" vertices by Klee [324] (see the next section!), and in "low" dimensions (for $d \leq 8$); see Grünbaum [252, p.175]. You may observe that the result is quite easily derived from the Dehn-Sommerville equations for $d \leq 5$.

Finally, in 1970 McMullen gave a complete proof of the upper-bound conjecture — since then it has been known as the upper bound theorem. McMullen's proof is amazingly simple and elegant, combining two key tools: shellability and h-vectors. This section presents this proof of the upper bound theorem, following McMullen's original paper [389]. Here is the theorem.

Theorem 8.23 (Upper bound theorem). (McMullen [389])
If P is a d-polytope with $n = f_0$ vertices, then for every k it has at most as many k-faces as the corresponding cyclic polytope (cf. Example 0.6):

$$f_{k-1}(P) \quad \leq \quad f_{k-1}(C_d(n)).$$

Here equality for some k with $\lfloor \frac{d}{2} \rfloor \leq k \leq d$ implies that P is neighborly.

The first fact to note is that we can restrict our attention to simplicial polytopes.

Lemma 8.24. (Klee [324] and McMullen [387]) *The vertices of a d-polytope P can be perturbed in such a way that the resulting polytope P' (with the same number of vertices) is simplicial, and*

$$f_{k-1}(P) \quad \leq \quad f_{k-1}(P')$$

for $0 \leq k \leq d$. Here equality for some $k > \lfloor \frac{d}{2} \rfloor$ can occur only if P is simplicial.

This is not the hard part, so we avoid the *distraction* of a proof. Thus from now on we only consider simplicial d-polytopes: this is essential, because it allows us to use the Dehn-Sommerville equations! What do they get us?

First, we note that we always have

$$f_{k-1} \;\leq\; \binom{n}{k},$$

with equality if and only if P is k-neighborly. This bound is achieved with equality for $k \leq \lfloor \frac{d}{2} \rfloor$ in the case of neighborly polytopes like the cyclic polytopes of Example 0.6. For $k > \lfloor \frac{d}{2} \rfloor$ this bound cannot be achieved, except in the case of a simplex, by Exercise 0.10.

However, the face numbers $f_{-1}, f_0, \ldots, f_{\lfloor \frac{d}{2} \rfloor - 1}$ already determine the complete f-vector. Namely, they determine $h_0, \ldots, h_{\lfloor \frac{d}{2} \rfloor}$ by Definition 8.18, and the Dehn-Sommerville equations give us the rest of the h-vector. In particular, all neighborly simplicial d-polytopes with n vertices have the same f-vector as the cyclic polytope $C_d(n)$. In this context recall that for odd $d \geq 3$, neighborly polytopes need not be simplicial; however, by Lemma 8.24 the nonsimplicial ones have smaller f_{k-1} for $k > \lfloor \frac{d}{2} \rfloor$, and thus we don't worry about them here.

Let's look at the expression of the f-vector in terms of $h_0, \ldots, h_{\lfloor \frac{d}{2} \rfloor}$. To get the prettiest possible formulas, we will use the notation

$$\sum_{i=0}^{\frac{d}{2}}{}^{*}\, T_i \;=\; \begin{cases} T_0 + T_1 + \cdots + \ T_{\lfloor \frac{d}{2} \rfloor} & \text{if } d \text{ is odd,} \\ T_0 + T_1 + \cdots + \frac{1}{2} T_{\lfloor \frac{d}{2} \rfloor} & \text{if } d \text{ is even.} \end{cases}$$

That is, the asterisk means that we take only half of the last term for $i = \frac{d}{2}$ if d is even, and take the whole last term for $i = \lfloor \frac{d}{2} \rfloor = \frac{d-1}{2}$ if d is odd. Similarly, we will use $\sum_*$ to denote a sum where only half of the first term is taken if the starting index of the summation is integral.

For $k \geq \lfloor \frac{d}{2} \rfloor$, we have with this notation

$$\begin{aligned} f_{k-1} &= \sum_{i=0}^{d} \binom{d-i}{k-i} h_i \qquad \text{(where the terms vanish for } i > k\text{)} \\ &= \sum_{i=0}^{\frac{d}{2}}{}^{*} \binom{d-i}{k-i} h_i \;+\; \sum_{i=\frac{d}{2}}^{d}{}_{*} \binom{d-i}{k-i} h_i \\ &= \sum_{i=0}^{\frac{d}{2}}{}^{*} \left(\binom{d-i}{k-i} + \binom{i}{k-d+i} \right) h_i, \end{aligned} \tag{8.25}$$

where for the last equality we have substituted $d - i$ for i, and used the Dehn-Sommerville equations.

Looking at that, we see that what we "really have to prove" is that for $k \leq \lfloor \frac{d}{2} \rfloor$ the neighborly simplicial polytopes not only maximize f_{k-1} (as we know), but they also maximize h_k. That is, the following lemma is "more than enough."

Lemma 8.26. [389, Lemma 2] *Let P be a simplicial d-polytope on $f_0 = n$ vertices. Then for $0 \leq k \leq d$,*

$$h_k(P) \;\leq\; \binom{n-d-1+k}{k}.$$

Equality holds for all k with $0 \leq k \leq l$ if and only if $l \leq \lfloor \frac{d}{2} \rfloor$ and P is l-neighborly.

The statement and proof of this lemma are the key steps in McMullen's solution of the upper-bound conjecture [389] (with the notation $g_{k-1}^{(d)}$ for what we now call h_k).

Proof. The proof is done by induction on k. The lemma is clearly true for $k = 0$, since we have defined h_0 to be 1. Thus it suffices to verify

$$\frac{h_{k+1}}{h_k} \quad \leq \quad \frac{\binom{n-d+k}{k+1}}{\binom{n-d-1+k}{k}},$$

that is,

$$(k+1)h_{k+1} \quad \leq \quad (n-d+k)h_k \tag{8.27}$$

for $k \geq 0$.

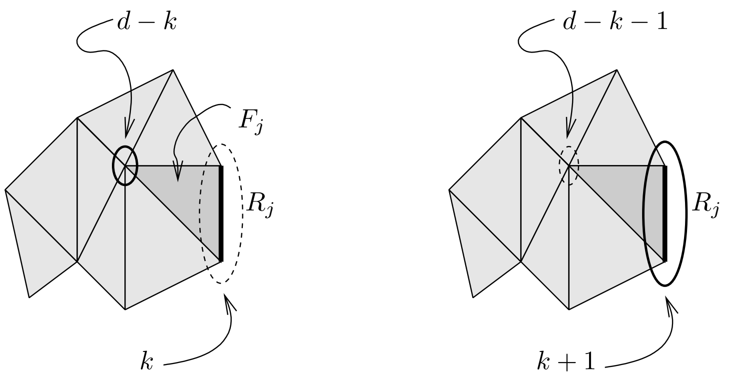
We get this by putting together two parts. The first one is the formula

$$\sum_{v\in\mathrm{vert}(\mathcal{C})} h_k(\mathcal{C}/v) \quad = \quad (k+1)h_{k+1}(\mathcal{C}) \;+\; (d-k)h_k(\mathcal{C}), \tag{8.27a}$$

where we use $\mathcal{C}/v$ as a convenient abbreviation for the link of v in the simplicial complex $\mathcal{C}$, that is, $\mathcal{C}/v := \mathrm{link}(v, \mathcal{C})$.

Equation 8.27a is easy to prove, because it is valid also during a shelling, when instead of $\mathcal{C} = \mathcal{C}(\partial P)$ we consider $\mathcal{C}_j := \mathcal{C}(F_1 \cup \cdots \cup F_j)$, and because a shelling on ∂P also induces a shelling order for all the links, by Lemma 8.7. The formula is valid at the beginning, for the empty complex $\mathcal{C}_0$ (when no facet is present, and all terms vanish). Now assume a new facet F_j is added, and consider its contribution to $\sum_v h_k(\mathcal{C}/v)$. Clearly this only affects the terms for vertices $v \in F_j$. There are two cases.

If $v \notin R_j$, then there is a new face of size $|R_j|$ in the link of v. This affects $h_k(\mathcal{C}/v)$ only if $|R_j| = k$, and in this case we get a contribution of "1" to $|F_j \backslash R_j| = d - k$ different summands. (The left drawing in our sketch indicates this case, for $k = 1$ and $d = 3$.) In this case, we also get that $h_k(\mathcal{C})$ increases by 1, and thus the right-hand side increases by $d-k$, as it should.



If $v \in R_j$, then we get a minimal new face of size $|R_j|-1$ in the link of v. So we get a contribution to $h_k(\mathcal{C}/v)$ only if $|R_j| = k+1$, and in this case we get a contribution of "1" to $k+1$ different terms on the left-hand side of the equation. At the same time, we get that $h_{k+1}(\mathcal{C})$ increases by 1, so the right-hand side increases by $k+1$, and we are even. (The right drawing in our sketch depicts this case.)

This proves equation (8.27a).

The second part we need is an inequality,

$$\sum_{v \in \mathrm{vert}(\mathcal{C})} h_k(\mathcal{C}/v) \quad \leq \quad n\, h_k(\mathcal{C}). \tag{8.27b}$$

For this, we prove that $h_k(\mathcal{C}/v) \leq h_k(\mathcal{C})$ holds for all n vertices $v \in \mathrm{vert}(\mathcal{C})$. To see this, take a shelling that shells the star of v first. This means that the minimal new face in $\mathcal{C}$ and in the link $\mathcal{C}/v$ coincide at every step while we are shelling the star. Later, after the shelling has left the star, we may get new contributions to $h_k(\mathcal{C})$, but not any more to $h_k(\mathcal{C}/v)$. With this we get the inequality (8.27b), and putting it together with equation (8.27a) we derive the inequality (8.27).

What about the equality case? To get $h_k(\mathcal{C}/v) = h_k(\mathcal{C})$, it is necessary that in a shelling that starts with the star of v, there is no "new" face of size at most k outside the star of v. Thus we get that, for $l \geq 1$, the equality $h_k(\mathcal{C}/v) = h_k(\mathcal{C})$ holds for all $k \leq l$ if and only if in a shelling that starts with the star of v, there is no minimal new face of size at most l outside the star of v. Equivalently, this says that every face G with at most $|G| \leq l$ vertices is contained in the star of v, so that $G \cup \{v\}$ is a face, too. Equality in (8.27b) holds only if we have equality for all vertices v. From this we get that equality in (8.27b), and thus in (8.27), holds for all $k \leq l$ if and only if $\mathcal{C}$ is $(l+1)$-neighborly. $\square$

On the way, we have also computed the f-vector of the neighborly polytopes: for this we only have to put the equality case of Lemma 8.26 into the formula 8.25.

Corollary 8.28. *If P is a simplicial neighborly d-polytope with $f_0 = n$ vertices, then*

$$f_{k-1} \quad = \quad \sum_{i=0}^{\frac{d}{2}}{}^{*} \left(\binom{d-i}{k-i} + \binom{i}{k-d+i} \right) \binom{n-d-1+i}{i}$$

for $0 \leq k \leq d$. For every k this gives the maximal number of $(k-1)$-faces for a d-polytope with n vertices.

For $k = d$, this reduces to a formula for the number of facets of the cyclic polytope $C_d(n)$:

$$f_{d-1} \quad = \quad \sum_{i=0}^{\frac{d}{2}}{}^{*}\, 2\binom{n-d-1+i}{i}.$$

(Compare this to Exercise 0.9!) Note that in fixed dimension, $f_{d-1}(C_d(n))$ grows like a polynomial of degree $\lfloor \frac{d}{2} \rfloor$ in the number of vertices.

Here is a brief asymptotic argument, due to Seidel [490] (see also Mulmuley [417]), for this corollary to the upper bound theorem. Namely, consider any shelling of ∂P. For every facet we get that either the restriction R_j or its complement $F_j \backslash R_j$ has size at most $\lfloor \frac{d}{2} \rfloor$. So, either in the shelling or in its reverse we have that F_j has a restriction of size at most $\lfloor \frac{d}{2} \rfloor$, and the restriction sets in a shelling are distinct by construction. Thus the number of facets is at most twice the number of k-faces of P with $k \le \lfloor \frac{d}{2} \rfloor$. From this we get

$$f_{d-1} \quad \le \quad 2 \sum_{i=0}^{\lfloor \frac{d}{2} \rfloor} \binom{n}{i},$$

and this rough estimate bounds f_{d-1} by a polynomial of degree $\lfloor \frac{d}{2} \rfloor$ in n.

8.5 Some Extremal Set Theory

We have used already that the simplicial complex $\mathcal{C}$ with n vertices can be identified with a *set system*, the collection of subsets $\mathcal{S}(\mathcal{C})$ of an n-set,

$$\mathcal{S}(\mathcal{C}) \quad := \quad \{\mathrm{vert}(G) : G \in L(\mathcal{C})\}.$$

For the following we identify the vertex set of $\mathcal{C}$ with the set

$$[n] \quad := \quad \{1, 2, \ldots, n\},$$

and the k-faces of the complex with the $(k+1)$-subsets of the ground set $[n]$, for $-1 \le k \le \dim(\mathcal{C})$. Thus, if $\mathcal{C}$ is a pure $(d-1)$-dimensional simplicial complex, then it is determined by its family of facets, which is a subset of $\binom{[n]}{d}$, the collection of d-subsets of $[n]$.

The construction behind this identifies *geometric simplicial complexes*, as we get them for example as boundary complexes of simplicial polytopes, with *abstract simplicial complexes*, where we only retain the information on the vertex set, and the information about "which vertex sets correspond to faces, respectively facets." This approach is useful for all problems that are not concerned with the geometry of a complex, but only with its combinatorial structure. The combinatorial structure of a complex, however, is *faithfully* represented by the abstract set system: from the set system data, it is easy to reconstruct the simplicial complex (this is a process known as *geometric realization* of the abstract simplicial complex).

On the next page is a "generic picture" of a simplicial complex, viewed as a set system. (Of course, this need not be *your* way of viewing set systems — supply your own sketch!) The left version just shows you the "shape" of a simplicial complex within the lattice of all subsets of $[n]$, while the